Cross-Lingual Hallucination Drift in LLMs: Does It Depend on Task Type?

Midway Report

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Abstract

Large Language Models (LLMs) are known to hallucinate in non-English languages, but whether this cross-lingual drift is task-dependent remains an open question. We test this by measuring hallucination rates of Aya Expanse 8B across English, Spanish, and Swahili on two structurally different tasks: factual QA (TruthfulQA) and commonsense reasoning (XCOPA), using GPT-4o-mini as an LLM-as-a-Judge evaluator. Our completed experiments yield a striking result: hallucination drift is strongly task-dependent. On TruthfulQA, Spanish drift is negligible ($\Delta\text{HR} = -2.66$ pp, $p = 0.60$). On XCOPA, Swahili drift is catastrophic ($\Delta\text{HR} = +90.67$ pp, $p < 0.001$). The cross-task drift interaction score $\Phi = -93.33$ pp is statistically significant ($\chi^2 = 170.62$, $p < 0.001$). A secondary finding is that when hallucinating on XCOPA in Swahili, the model generates nearly twice as many tokens as on faithful responses (186 vs. 100), suggesting incoherent rambling rather than confident fabrication. These results extend and refine prior work by showing that task type is a major confound in cross-lingual hallucination estimation.

1 Introduction

Hallucination in LLMs, generating content that is factually incorrect or unfaithful is a well-documented failure mode (Alansari et al., 2025). A natural concern as LLMs are deployed globally is whether hallucination rates are higher in non-English languages, particularly low-resource ones where training data is sparse (Blasi et al., 2022). This phenomenon is termed *Cross-Lingual Hallucination Drift*: the change in hallucination rate when a model operates in a non-English language.

A pivotal study by ul Islam et al. (2025) measured token-level hallucination rates across 30 languages and 6 open-source LLM families, finding that *length-normalized hallucination rates are uncorrelated with language resource level*, a surprising result that overturns the common assumption that low-resource languages hallucinate more. However, that study evaluated a single task type: open-domain, knowledge-intensive long-form QA. Whether this finding holds across structurally different task types remains unexplored.

We investigate this gap. Factual recall (TruthfulQA) requires the model to retrieve internally stored parametric knowledge, while commonsense reasoning (XCOPA) requires applying world knowledge to select a plausible cause or effect (Qi et al., 2023). These two task types may expose fundamentally different failure modes across languages. We formally define the **Drift Interaction Score**:

$$\Phi_l = \Delta\text{HR}_{l,\,\text{TruthfulQA}} - \Delta\text{HR}_{l,\,\text{XCOPA}}$$

where $\Delta\text{HR}_{l,t} = \text{HR}_{l,t} - \text{HR}_{\text{en},t}$. A large $|\Phi_l|$ indicates drift is task-dependent for language l. Our completed experiments confirm this hypothesis decisively.

2 Methods

2.1 Datasets

We use two benchmarks representing structurally different task types.

TruthfulQA (Lin et al., 2022) consists of 817 questions across 38 domains designed to elicit confident false beliefs. We use the English original and the Spanish translation from the `alexandrainst/m_truthfulqa` HuggingFace dataset. Spanish is a mid-resource language with substantial LLM training data coverage.

XCOPA is a cross-lingual commonsense causal reasoning benchmark spanning 11 typologically diverse languages. Each example presents a premise and two candidate causes or effects; the model must select the more plausible one. We use the English split from SuperGLUE COPA and the Swahili (`sw`) split. Swahili is a low-resource language. Spanish is not available in XCOPA, so our cross-task comparison uses Spanish for TruthfulQA and Swahili for XCOPA.

We sample 150 examples per language per task (random seed 42), yielding 600 total evaluation instances across four language–task cells.

2.2 Models

Target model. We evaluate **Aya Expanse 8B**, a multilingual open-source instruction-tuned model by Cohere Labs, queried with zero-shot prompts written in the target language for each cell. Prompts are templated separately for TruthfulQA (free-form answer) and XCOPA (choose A or B).

Judge model. We use **GPT-4o-mini** as our LLM-as-a-Judge evaluator, following the methodology of Kang et al. (2024). The judge receives each (question, model response) pair and labels it *Hallucinated* or *Faithful* with a brief justification. We chose GPT-4o-mini for its strong multilingual comprehension across all three languages.

2.3 Evaluation Pipeline

The pipeline proceeds in five stages: (1) dataset download and sampling (2) zero-shot inference with Aya Expanse 8B on all four language–task cells(3) GPT-4o-mini judge labeling, with automatic retry for any `ERROR` labels (4) hallucination rate computation and (5) statistical significance testing.

We compute the hallucination rate as:

$$\mathrm{HR}_{l,t} = \frac{\#\,\text{Hallucinated}}{N_{l,t}} \times 100\,(\%)$$

where $N_{l,t} = 150$ for all cells. All code is publicly available in our repository.

3 Experiments

3.1 Setup

All experiments run on Python 3.10 with `pandas`, `scipy`, `matplotlib`, and the HuggingFace `datasets` library. Aya Expanse 8B inference requires a GPU, judge calls use the OpenAI API. Total labeled examples: 600 (150 per cell). Judge labels were inspected and retried for any malformed outputs.

3.2 Statistical Tests

We apply three tests to assess significance of hallucination differences:

Task	Language	HR (%)	ΔHR (pp)
TruthfulQA	English	27.33	—
TruthfulQA	Spanish	24.67	−2.66
XCOPA	English	8.00	—
XCOPA	Swahili	98.67	+90.67

Table 1: Hallucination rates (HR) and drift (ΔHR vs. English baseline) per language–task cell. $N = 150$ per cell.

- **Mann-Whitney U** (per task): tests whether hallucination rates differ between English and the non-English language within a task.
- **Chi-squared** (2×2 contingency): tests whether hallucination outcome is independent of task type for non-English responses, directly testing the Φ hypothesis.
- **Fisher's exact**: robustness check on the same 2×2 table, preferred for smaller cell counts.

4 Results

4.1 Hallucination Rates and Drift

Table 1 shows hallucination rates across all four language–task cells. TruthfulQA rates are comparable between English (27.33%) and Spanish (24.67%), yielding a negligible and non-significant drift of -2.66 pp. In contrast, XCOPA Swahili hallucination is near-total (98.67%), representing a drift of $+90.67$ pp above the English XCOPA baseline (8.00%).

4.2 Drift Interaction and Statistical Significance

Table 2 presents all statistical test results. The XCOPA en→sw drift is highly significant ($p < 0.001$), while TruthfulQA en→es drift is not ($p = 0.60$). The cross-task χ^2 test confirms that hallucination outcome is not independent of task type ($\chi^2 = 170.62$, $p < 0.001$), validated by Fisher's exact test (OR $= 226.0$, $p < 0.001$). The aggregate drift interaction score is $\Phi = -93.33$ pp, far exceeding our pre-specified minimum effect threshold of $|\Phi| \geq 2$ pp.

4.3 Token Verbosity

A secondary finding: when hallucinating on XCOPA in Swahili, Aya generates an average of 186 tokens per response, compared to 100 tokens for faithful responses in the same cell, an 86% increase. This pattern does not hold for TruthfulQA in either language, suggesting that XCOPA Swahili

Test	Comparison	Stat.	p	Sig.
Mann-Whitney U	TruthfulQA: en vs es	11550.0	0.600	No
Mann-Whitney U	XCOPA: en vs sw	1050.0	<0.001	Yes
Chi-squared	es/TQA vs sw/XCOPA	170.62	<0.001	Yes
Fisher's exact	es/TQA vs sw/XCOPA	OR=226.0	<0.001	Yes

Table 2: Statistical test results. TQA = TruthfulQA. Chi-squared and Fisher's exact test the Φ hypothesis directly.

hallucinations reflect *incoherent rambling* rather than the confident but wrong short answers typical of factual QA hallucination.

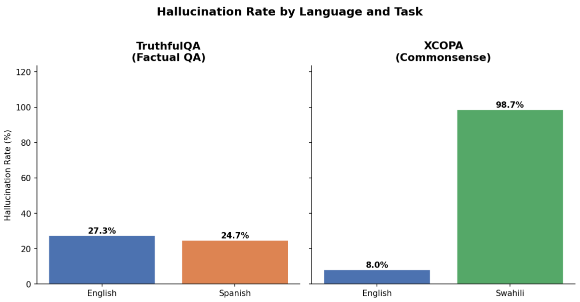

Figure 1: Hallucination rates by language and task. XCOPA Swahili is near-total (98.67%), while TruthfulQA drift between English and Spanish is negligible.

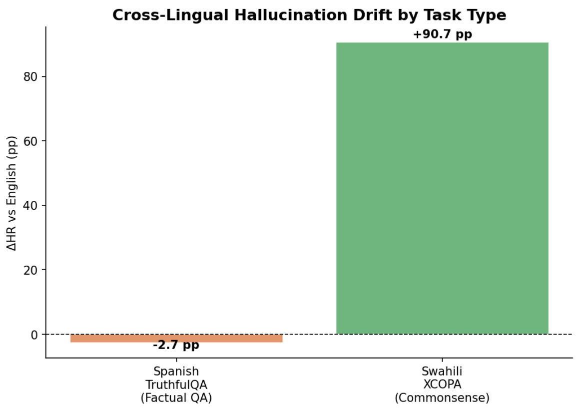

Figure 2: Cross-lingual drift (ΔHR vs. English) by language–task cell. Spanish TruthfulQA shows near-zero drift (-2.7 pp), while Swahili XCOPA shows catastrophic drift ($+90.7$ pp).

4.4 Discussion

Our results provide strong evidence that cross-lingual hallucination drift is task-dependent. The contrast between tasks is stark: a multilingual model that handles Spanish factual QA nearly as well as English collapses almost entirely on Swahili commonsense reasoning. This suggests that commonsense reasoning in XCOPA may be more sensitive to linguistic and cultural grounding in the target language, the model cannot fall back on parametric factual recall and instead produces incoherent output. This finding directly extends ul Islam et al. (2025): their null correlation between HR and language resource level may hold for knowledge-intensive QA but not for commonsense reasoning.

One limitation is that Spanish and Swahili appear in different tasks (TruthfulQA and XCOPA, respectively), since Spanish is not available in XCOPA. Our Φ score therefore conflates task-type and language-identity differences. We address this by framing Φ as a cross-task aggregate comparison rather than a per-language score, and note it as a priority for the final report.

5 Project Plan

All core experiments are complete. The remaining work for the final report is:

1. **Qualitative error analysis.** Categorize GPT-4o-mini judge reasons (e.g., incoherent, wrong answer, fabricated, code-switching) and report hallucination type distributions per cell.

2. **XCOPA question-type breakdown.** Separate XCOPA hallucination rates by question type (*cause* vs. *effect*) to assess whether causal direction affects the failure mode.

3. **Dual-judge agreement.** Run a second judge on a 50-sample subset per cell to compute inter-annotator agreement (Cohen's κ) and validate GPT-4o-mini reliability on Swahili.

4. **Final report write-up.** Expand the methods and discussion sections, incorporate error analysis findings, and polish figures for submission.

Team Contributions

Member	Role	Contributions
Bhoomika Monthy Rajashekar	Dataset download (TruthfulQA EN/ES, XCOPA EN/SW),	HuggingFace pipeline setup, prompt formatting and JSON storage, GitHub repo setup and data branch push
Chun Hsu	Evaluation & Metrics	LLM-as-a-Judge prompt design; hallucination rate computation; ΔHR and Φ metric implementation
Devinn Chi	Experiments	Zero-shot inference with Aya Expanse 8B across all language task cells, output logging and token count tracking, GPT-4o-mini judge runs
Anagha P Krishna	Analysis & Writing	statistical tests (Mann-Whitney, chi-squared, Fisher's exact), visualizations, report write-up

Table 3: Team task distribution for the midway report.

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